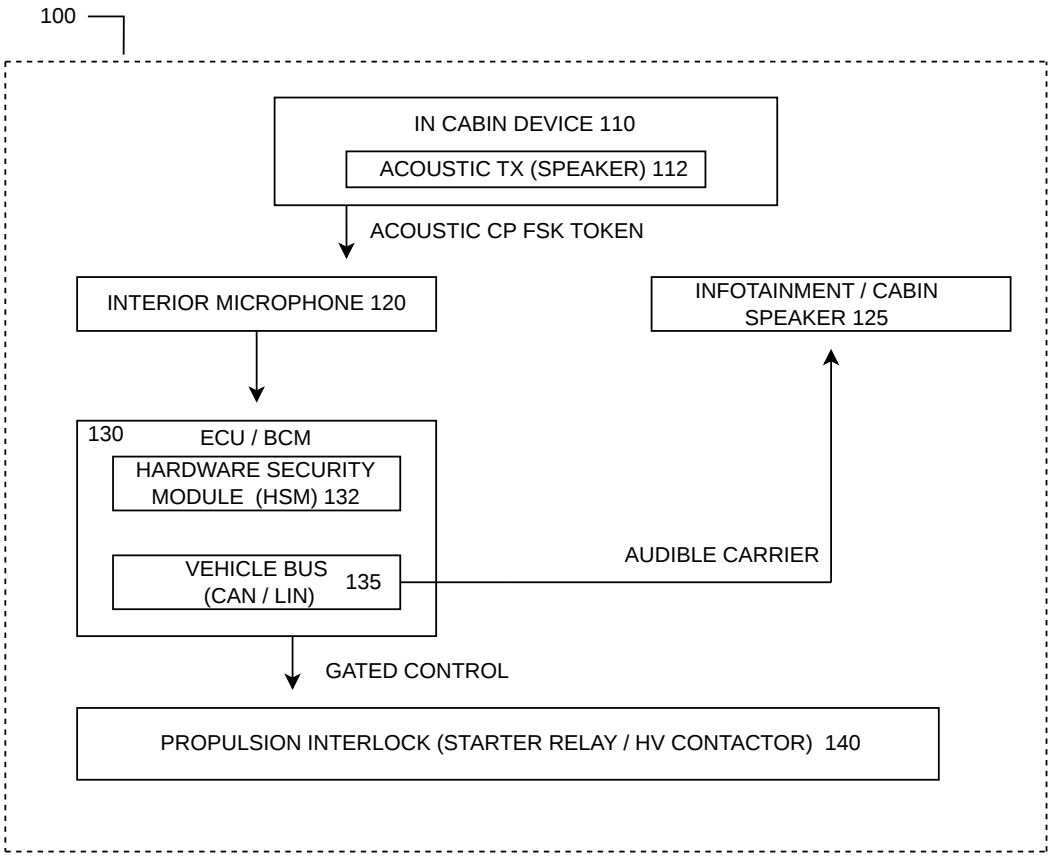
FIG 1/6



EXTERNAL THREAT SPACE

PRIMARY WIRELESS KEY 150

RF / BLE KEY FOB

EXTERNAL RELAY ATTACKER / EAVESDROPPER 160

ATTEMPTED RF/AUDIO RELAY DEVICE

FIG 2/6

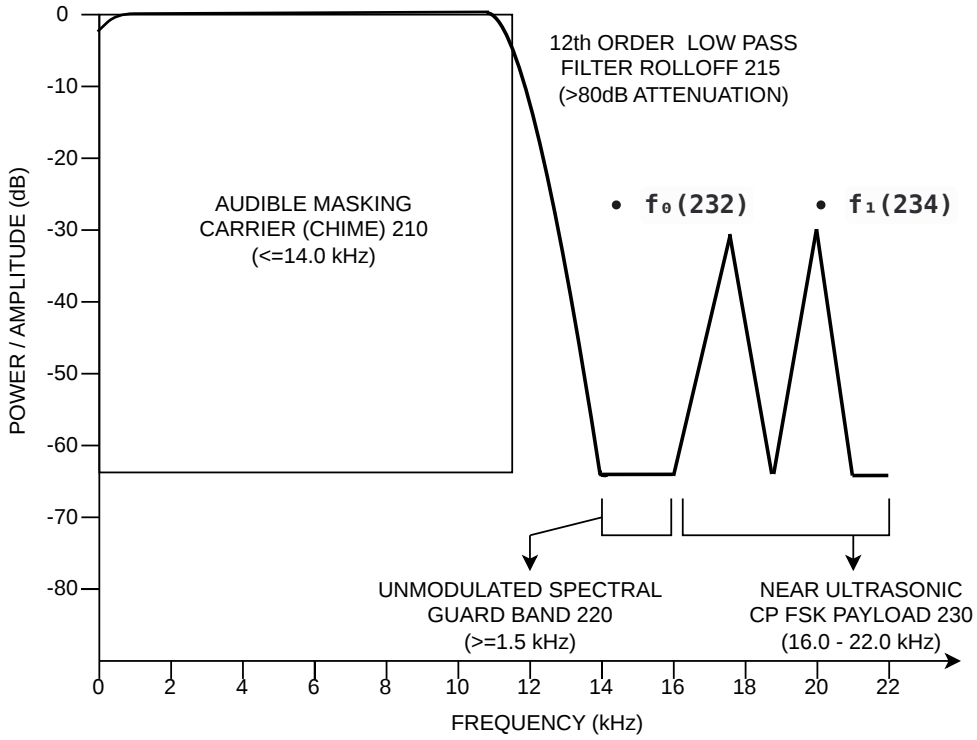

0
-10
-20
-30
-40
-50
-60
-70
-80
POWER / AMPLITUDE (dB)
12th ORDER LOW PASS
FILTER ROLLOFF 215
(>80dB ATTENUATION)
AUDIBLE MASKING
CARRIER (CHIME) 210
(<=14.0 kHz)
• f0(232)
• f1(234)
UNMODULATED SPECTRAL
GUARD BAND 220
(>=1.5 kHz)
NEAR ULTRASONIC
CP FSK PAYLOAD 230
(16.0 - 22.0 kHz)
0
2
4
6
8
10
12
14
16
18
20
22
FREQUENCY (kHz)

FIG 3/6

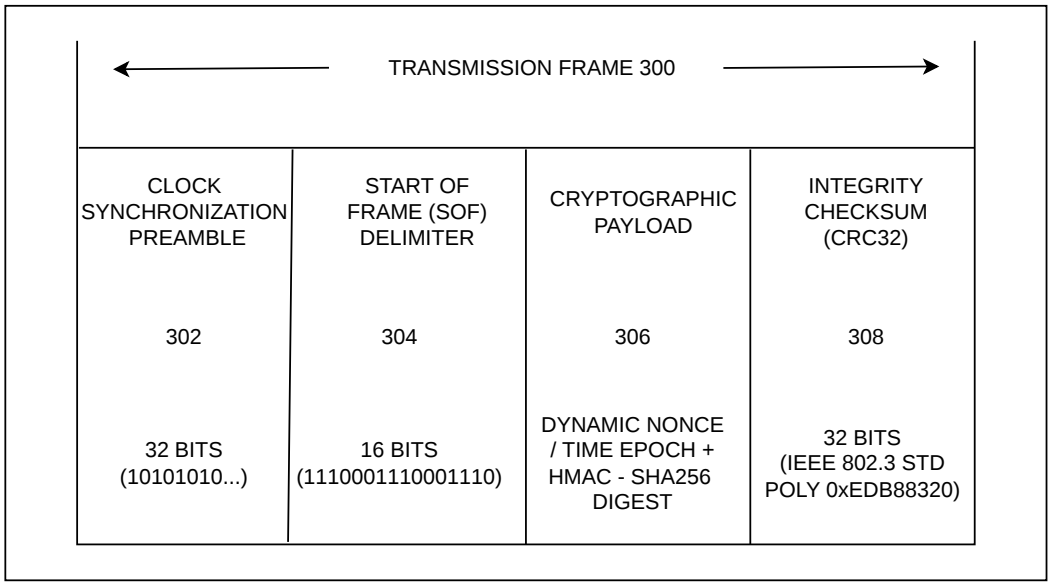

TRANSMISSION FRAME 300
CLOCK SYNCHRONIZATION PREAMBLE
START OF FRAME (SOF) DELIMITER
CRYPTOGRAPHIC PAYLOAD
INTEGRITY CHECKSUM (CRC32)
302
304
306
308
32 BITS (10101010...)
16 BITS (1110001110001110)
DYNAMIC NONCE / TIME EPOCH + HMAC - SHA256 DIGEST
32 BITS (IEEE 802.3 STD POLY 0xEDB88320)

FIG 4/6

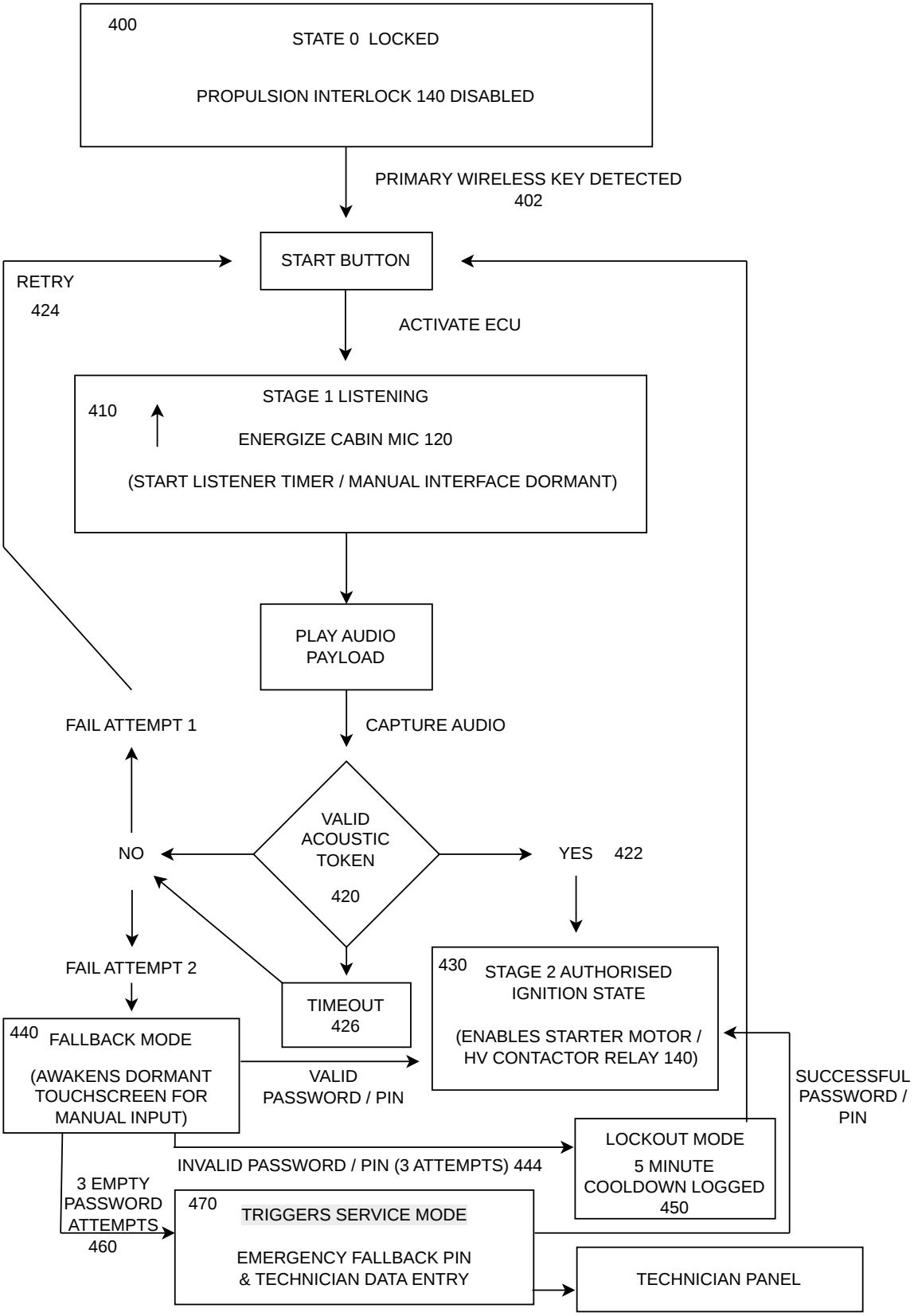

400
STATE 0 LOCKED
PROPULSION INTERLOCK 140 DISABLED
PRIMARY WIRELESS KEY DETECTED
402
START BUTTON
RETRY
424
ACTIVATE ECU
STAGE 1 LISTENING
410
ENERGIZE CABIN MIC 120
(START LISTENER TIMER / MANUAL INTERFACE DORMANT)
PLAY AUDIO PAYLOAD
FAIL ATTEMPT 1
CAPTURE AUDIO
VALID ACOUSTIC TOKEN
420
NO
YES 422
FAIL ATTEMPT 2
TIMEOUT
426
430
STAGE 2 AUTHORISED IGNITION STATE
(ENABLES STARTER MOTOR / HV CONTACTOR RELAY 140)
440
FALLBACK MODE
(AWAKENS DORMANT TOUCHSCREEN FOR MANUAL INPUT)
VALID PASSWORD / PIN
SUCCESSFUL PASSWORD / PIN
LOCKOUT MODE
5 MINUTE COOLDOWN LOGGED
450
INVALID PASSWORD / PIN (3 ATTEMPTS) 444
3 EMPTY PASSWORD ATTEMPTS
460
470
TRIGGERS SERVICE MODE
EMERGENCY FALLBACK PIN & TECHNICIAN DATA ENTRY
TECHNICIAN PANEL

FIG 5/6

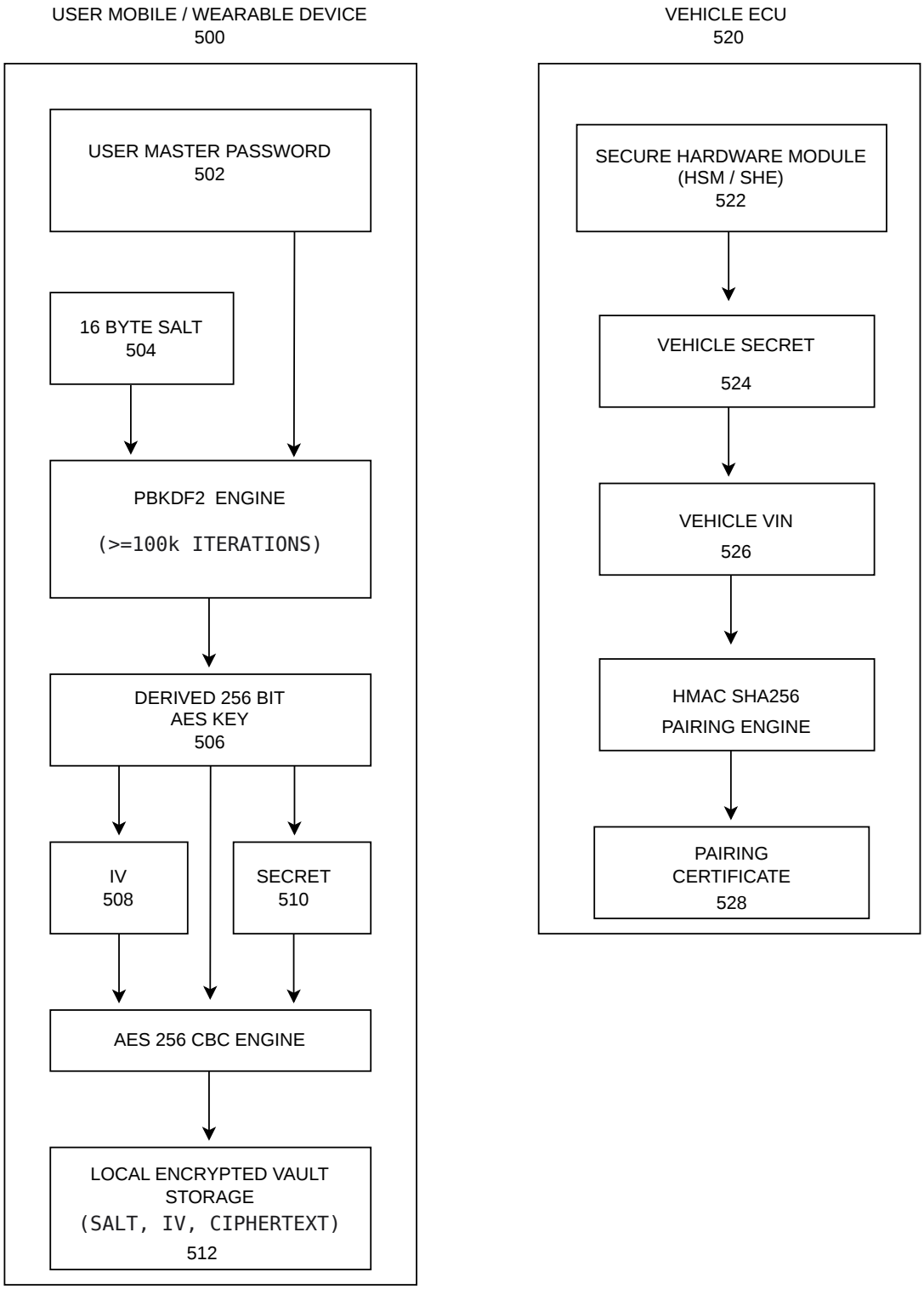
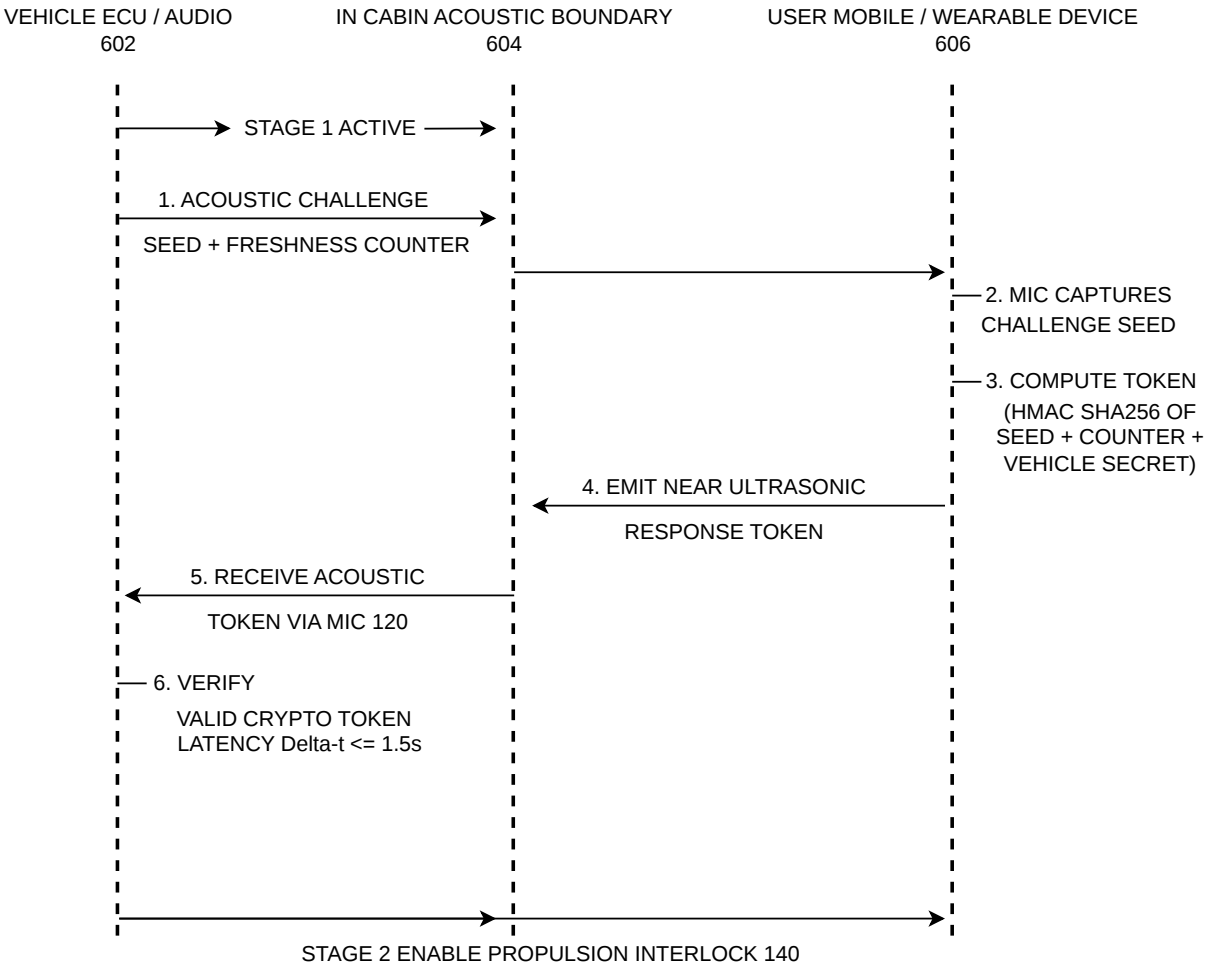

USER MOBILE / WEARABLE DEVICE
500
USER MASTER PASSWORD
502
16 BYTE SALT
504
PBKDF2 ENGINE
(>=100k ITERATIONS)
DERIVED 256 BIT
AES KEY
506
IV
508
SECRET
510
AES 256 CBC ENGINE
LOCAL ENCRYPTED VAULT
STORAGE
(SALT, IV, CIPHERTEXT)
512
VEHICLE ECU
520
SECURE HARDWARE MODULE
(HSM / SHE)
522
VEHICLE SECRET
524
VEHICLE VIN
526
HMAC SHA256
PAIRING ENGINE
PAIRING
CERTIFICATE
528

FIG 6/6


VEHICLE ECU / AUDIO
602
IN CABIN ACOUSTIC BOUNDARY
604
USER MOBILE / WEARABLE DEVICE
606
STAGE 1 ACTIVE
1. ACOUSTIC CHALLENGE
SEED + FRESHNESS COUNTER
2. MIC CAPTURES
CHALLENGE SEED
3. COMPUTE TOKEN
(HMAC SHA256 OF
SEED + COUNTER +
VEHICLE SECRET)
4. EMIT NEAR ULTRASONIC
RESPONSE TOKEN
5. RECEIVE ACOUSTIC
TOKEN VIA MIC 120
6. VERIFY
VALID CRYPTO TOKEN
LATENCY Delta-t <= 1.5s
STAGE 2 ENABLE PROPULSION INTERLOCK 140